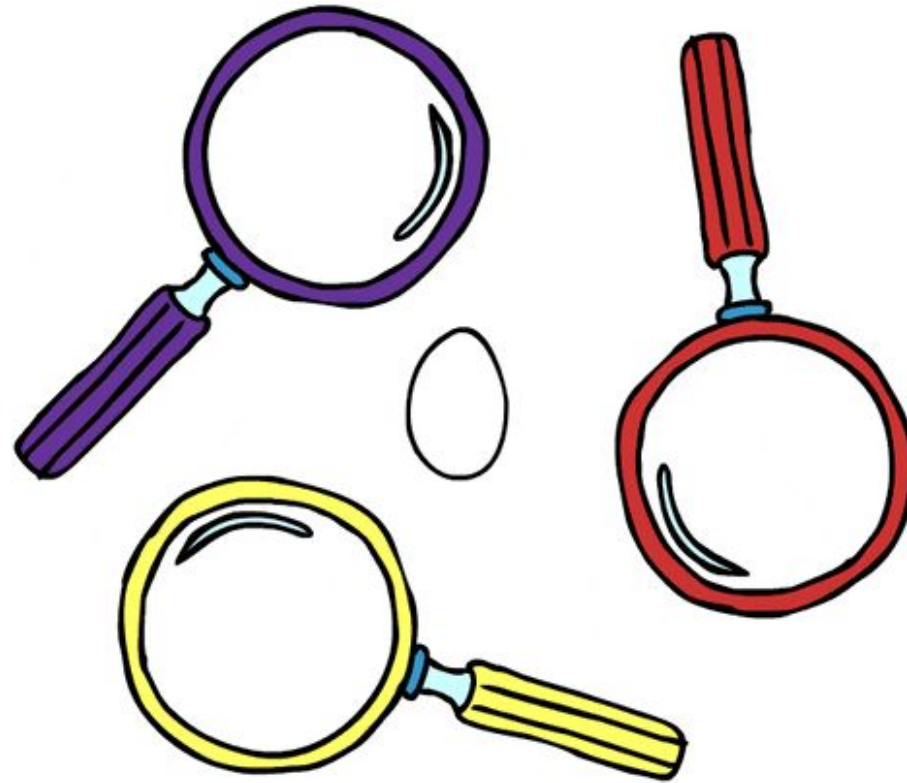


Research Methodology-II

Qualitative Research



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Which qualitative method focuses on personal stories?

Answer: Narrative

What is the sampling method commonly used in qualitative research?

Answer: Purposive

Which method involves group discussions led by a moderator?

Answer: Focus group

Which qualitative method is used to understand cultural practices?

Answer: Ethnography

Which analysis technique is used to identify patterns in qualitative data?

Answer: Thematic

What kind of questions does qualitative research typically use?

Answer: Open-ended

Which qualitative method involves watching behaviors in real settings?

Answer: Observation

What is the researcher's role in qualitative research?

Answer: Interpreter

What type of research design studies one specific case in depth?

Answer: Case Study



Objectives



At the end of the lecture student should be able to:

- Explain the types of data collected in qualitative research (e.g., words, narratives, open-ended responses).
- Understand the concept of thematic analysis and how it is used to interpret qualitative data.
- Appreciate the role of qualitative research in exploring patient behavior, attitudes, and barriers to dental care.
- Recognize the importance of context and subjectivity in qualitative research findings.
- Discuss ethical issues involved in qualitative research, such as informed consent, confidentiality, and sensitivity to participants' experiences.
- Value the contribution of qualitative research in enhancing patient-centered care and improving communication in dental practice.

A group of researchers worked in a village in South Sudan. They did a survey of villagers and asked them what were the most serious problems. The villagers said that one of the most serious problems was malaria.

How can we confirm that?

The researchers tested the blood of some of the villagers and found that malaria was very common. They gave people in the village bed-nets soaked in insecticide and told them that using the nets would decrease malaria.

The people started using the nets. The research team tested the blood of the people and found that there was a big reduction in malaria for those people who used the nets. So they felt the program was successful.

Later some other researchers came to the village. They found the people no longer used the nets. They said that they didn't work.

How can we know the reason?

The villagers were skeptical of outsiders and were less interested in cooperating in programs to reduce malaria.

In this case the reason the villagers stopped using the nets was a misunderstanding. By 'malaria' the researchers meant infection with the malaria parasite. But the villagers meant all fevers. Therefore, although the number of 'malaria infections' decreased, the total number of 'all fevers' did not change much in the community's eyes. For this reason, they felt the nets did not work against what the community called 'malaria' and abandoned using the nets.

TYPES OF QUALITATIVE RESEARCH

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graph TD; Center((TYPES OF QUALITATIVE RESEARCH)) --> Narrative((Narrative Method)); Center --> Phenomenological((Phenomenological Method)); Center --> GroundedTheory((Grounded Theory Method)); Center --> CaseStudy((Case Study Research)); Center --> Historical((Historical Method)); Center --> ActionResearch((Action Research)); Center --> FocusGroup((Focus Group)); Center --> Ethnographic((Ethnographic Research Method)); Narrative --> Phenomenological; Phenomenological --> GroundedTheory; GroundedTheory --> CaseStudy; CaseStudy --> Historical; Historical --> ActionResearch; ActionResearch --> FocusGroup; FocusGroup --> Ethnographic; Ethnographic --> Narrative;
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Uses storytelling to understand people's experiences

Researchers immerse themselves in a social setting to understand culture or behavior, beliefs, and practices

Studies lived experiences to gain deeper insights into how people understand those experiences

Gathering a small, carefully selected group of individuals to discuss a specific topic

A research method concerned with the generation of theory, which is 'grounded' in data that has been systematically collected and analysed

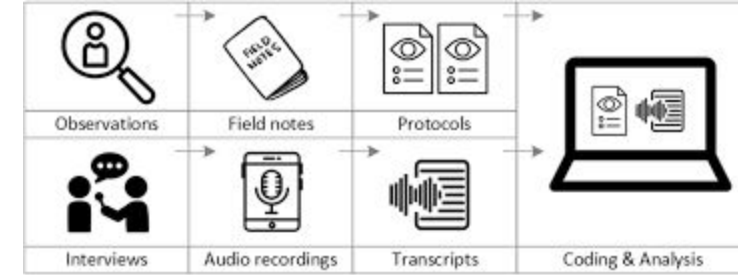
Actively engage in a cyclical process of identifying a problem, planning a course of action, implementing that action, observing the outcomes, and reflecting on the results to improve practices or solutions.

The focus is on a detailed examination of a specific case.

Systematically collecting and analyzing primary and secondary sources to understand past events

Qualitative Research: Data

Data is typically non-numerical and aims to explore concepts, meanings, or experiences in depth. Main types of data collected are:



1. Words Spoken or written language is the primary medium.

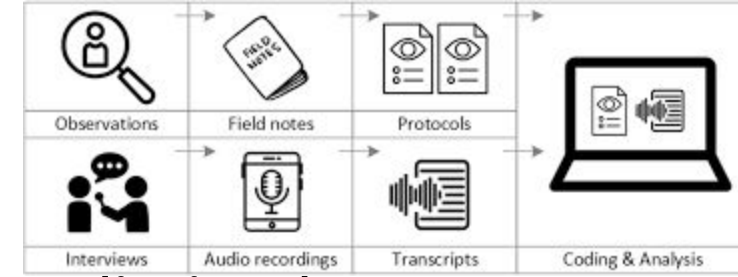
Examples: Interview transcripts, written reflections, responses to open-ended survey questions.

2. Narratives, Personal stories or accounts that provide insight into an individual's experiences, perceptions, and emotions.

Examples: Life histories, autobiographical accounts, patient stories in medical research.

3. Field Notes/Observational data recorded by researchers during or after fieldwork. Include both descriptive notes (what happened) and reflective notes (researcher's thoughts, interpretations).

Qualitative Research: Data

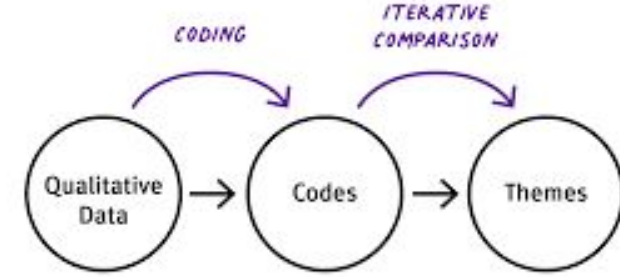


4. Documents and Texts Analysis of existing documents, such as diaries, letters, emails, official records, or social media posts. Used in content or discourse analysis. Example, A researcher conducts a content analysis of patient diaries written during recovery from oral surgery. The diaries include descriptions of pain levels, emotional responses, and coping strategies.

5. Visual and Audio Materials. While not "words" in the traditional sense, these are used in qualitative research to study behavior, environment, and context. Examples: Photographs, videos, audio recordings (interviews, conversations).

6. Artifacts Objects or tools that reflect cultural or social meaning. Often used in ethnographic research to understand practices and values. Example, in a study of dental hygiene practices in a rural community, researchers collect and analyze locally made toothbrushes (chew sticks) used by the residents.

Qualitative Research: Thematic Analysis



What is a “Theme”?

A theme is a pattern found in the data that captures something important or meaningful about the research question. It represents a level of response or meaning within the data set.

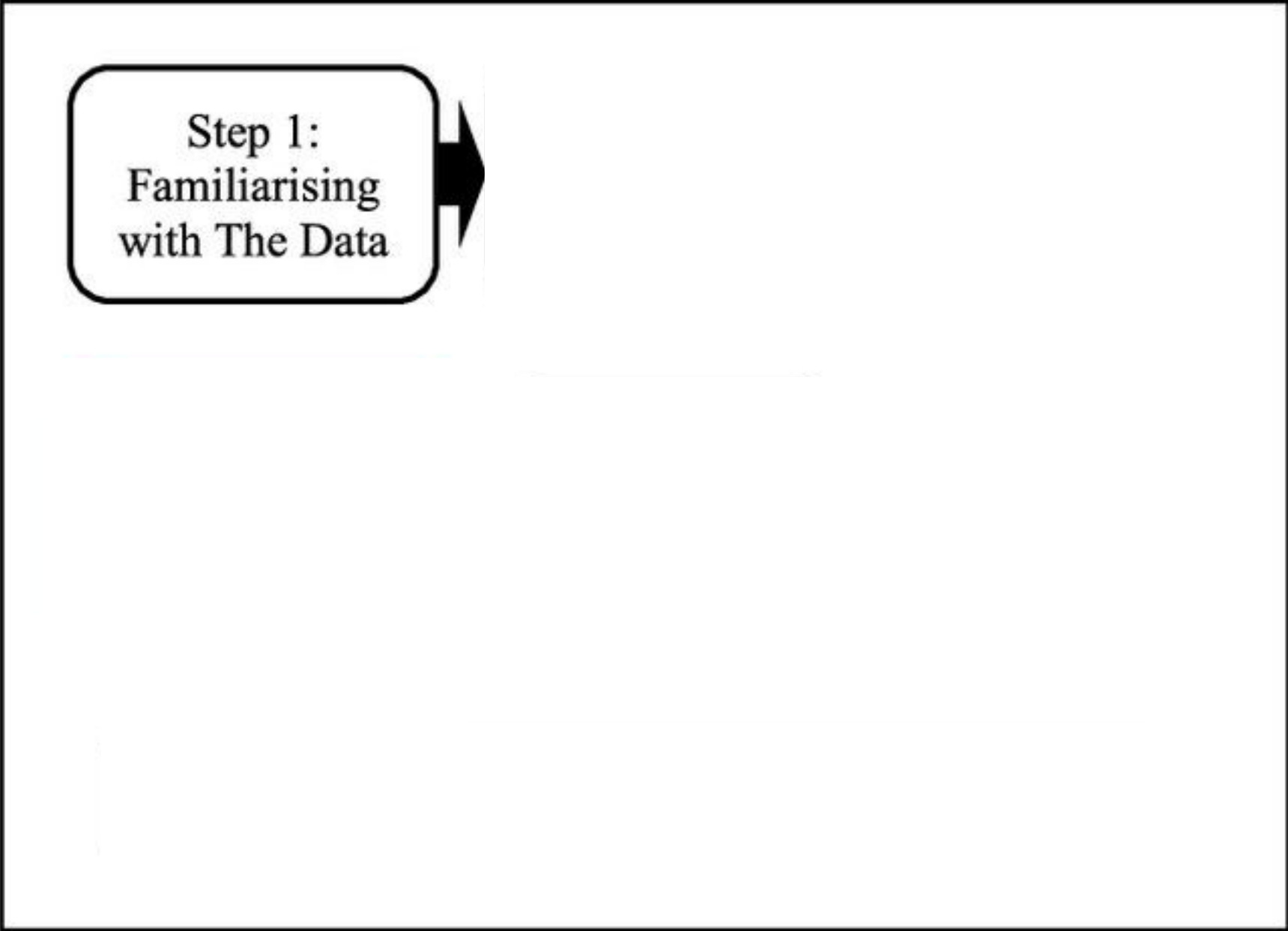
Thematic Analysis is a method used in qualitative research to identify, analyze, and interpret patterns within textual data.

Flexible: Can be used across different types of qualitative data (e.g., interviews, focus groups, open-ended survey responses).

Systematic: Follows a clear step-by-step process.

Rich and Detailed: Goes beyond counting words to examine the deeper meanings within the data.

Qualitative Research: Thematic Analysis



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graph LR; A[Step 1: Familiarising with The Data] --> B[Step 2: Generating Initial Themes]; B --> C[Step 3: Reviewing Themes]; C --> D[Step 4: Defining and Naming Themes]; D --> E[Step 5: Producing the Report];
```

Step 1:
Familiarising
with The Data

Steps in Thematic Analysis
(Commonly based on
Braun & Clarke, 2006)

Qualitative Research: Thematic Analysis

1. **Familiarization with Data:** Reading and re-reading transcripts. Noting initial impressions or observations.
2. **Generating Initial Codes:** Systematically labeling meaningful features of the data. Codes are like tags (e.g., "fear of diagnosis", "support from family").
3. **Searching for Themes:** Grouping codes into potential themes. For example, codes like "anxiety," "shock," and "uncertainty" might form a theme called Emotional Response to Illness.
4. **Reviewing Themes:** Checking if themes work in relation to the coded data and the entire dataset.
5. **Refining themes:** some may be combined, split, or discarded.
6. **Defining and Naming Themes:** Giving each theme a clear definition and name. Writing a description of what each theme is about.
7. **Producing the Report:** Writing up the analysis with data excerpts and interpretations. Relating the themes back to the research question and literature.

Research Question: What are the perceptions and experiences of first-year BDS students regarding the Introduction to Research course?

Themes Identified :

1. "Before this course, I had no idea what research really meant. Now I know how important it is."

Increased Awareness of Research: Students felt more informed about research principles and their relevance to dentistry.

2. "At first, I was totally lost when we discussed methodology. It felt like a different language!"

Initial Confusion and Anxiety: Many students felt overwhelmed by unfamiliar terms and the academic nature of research.

3. "It's good we learned this in first year. It makes later projects feel less intimidating."

Value of Early Exposure: Students appreciated being introduced to research early in their dental education.

4. "If we had a small project or case study, I'd understand better how to apply the concepts."

Need for Practical Examples: Students wanted more applied components such as hands-on projects or simpler examples.

5. "I never thought I'd be interested, but now I actually want to work on a research paper someday."

Motivation for Future Research :Some students felt inspired to pursue research in the future.

Interpretation

Thematic analysis revealed both challenges and benefits of the course.

Although some students initially struggled, they ultimately recognized the value of learning research early.

The findings suggest that incorporating more interactive and practical elements could enhance learning outcomes.

Summary of Findings (for report)

The Introduction to Research course was perceived as valuable but challenging. Students appreciated the early exposure to research but expressed a need for more interactive content.

Thematic analysis showed that the course increased awareness and motivation for future research despite initial confusion.

1- Increased Awareness of Research:

2- Initial Confusion and Anxiety:

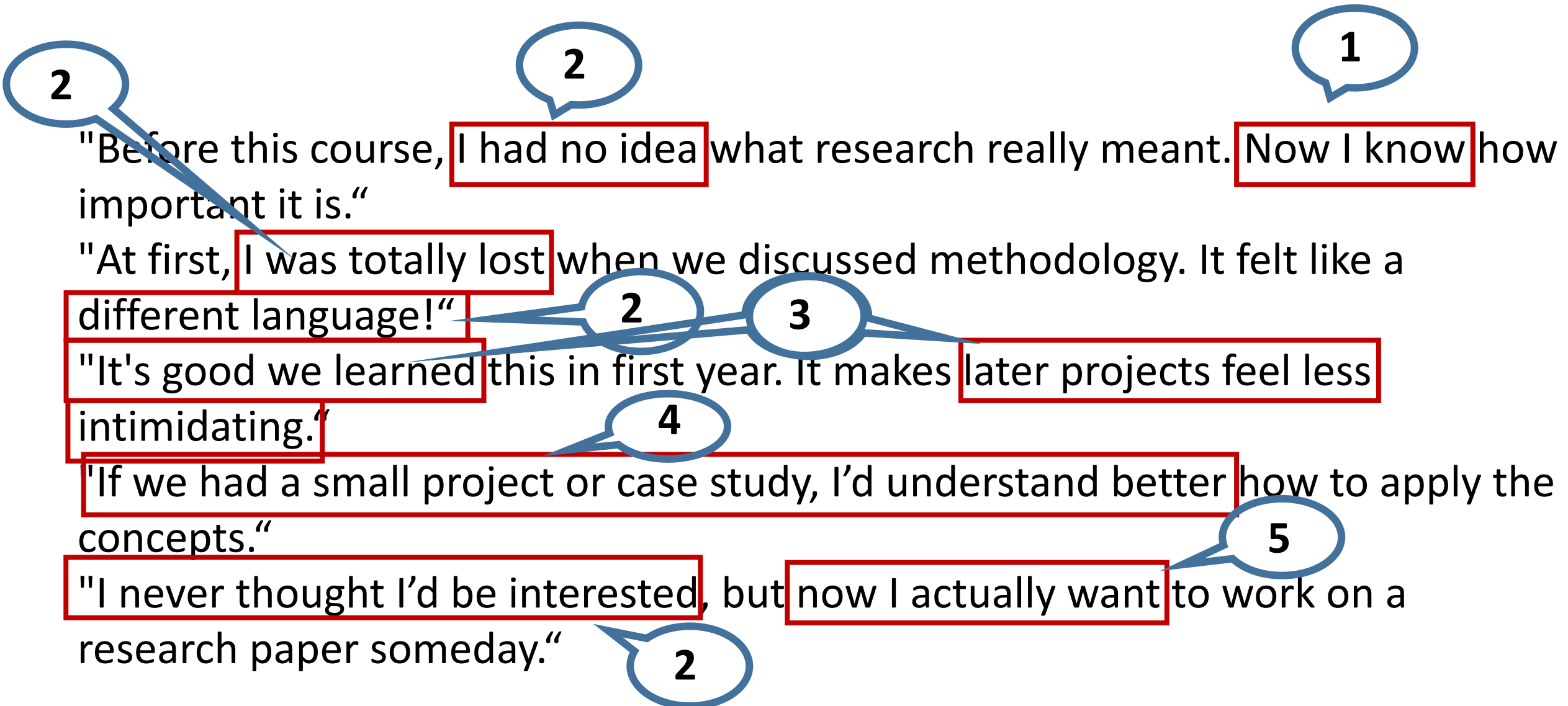
3- Value of Early Exposure:

4- Need for Practical Examples

5- Motivation for Future Research

Research Question: What are the perceptions and experiences of first-year BDS students regarding the Introduction to Research course?

Codes????



Qualitative Research: Role in Dentistry

Explains why patients behave a certain way

Explores Patient Attitudes Toward Dental Care

Identifies Barriers to Accessing Dental Care

Gives Voice to Under-represented Populations

Improves Patient-Centered Care

Example???

Qualitative Research: Context

What is Context?

Context refers to the background, environment, or setting in which a person's experiences or behaviors occur.

Cultural norms, Social structures, Personal circumstances, Time and place of interaction

The same behavior or response can mean very different things depending on the setting.

Ensures accurate interpretation: Without context, researchers may misinterpret or oversimplify complex human experiences.

Enhances credibility: Understanding where and how data were collected helps validate findings.

Example: A patient saying "I don't trust dentists"

Context A: The patient had a traumatic dental experience as a child.

Context B: The patient comes from a community with limited access to trained dental professionals.

Qualitative Research: Subjectivity

What is Subjectivity?

Subjectivity refers to the fact that people interpret and experience reality in personal, emotional, and unique ways

Captures human experience: Emotions, perceptions, beliefs, and values are central to understanding human behavior.

Acknowledges researcher influence: The researcher's background, position, and worldview may influence interpretation (researcher reflexivity).

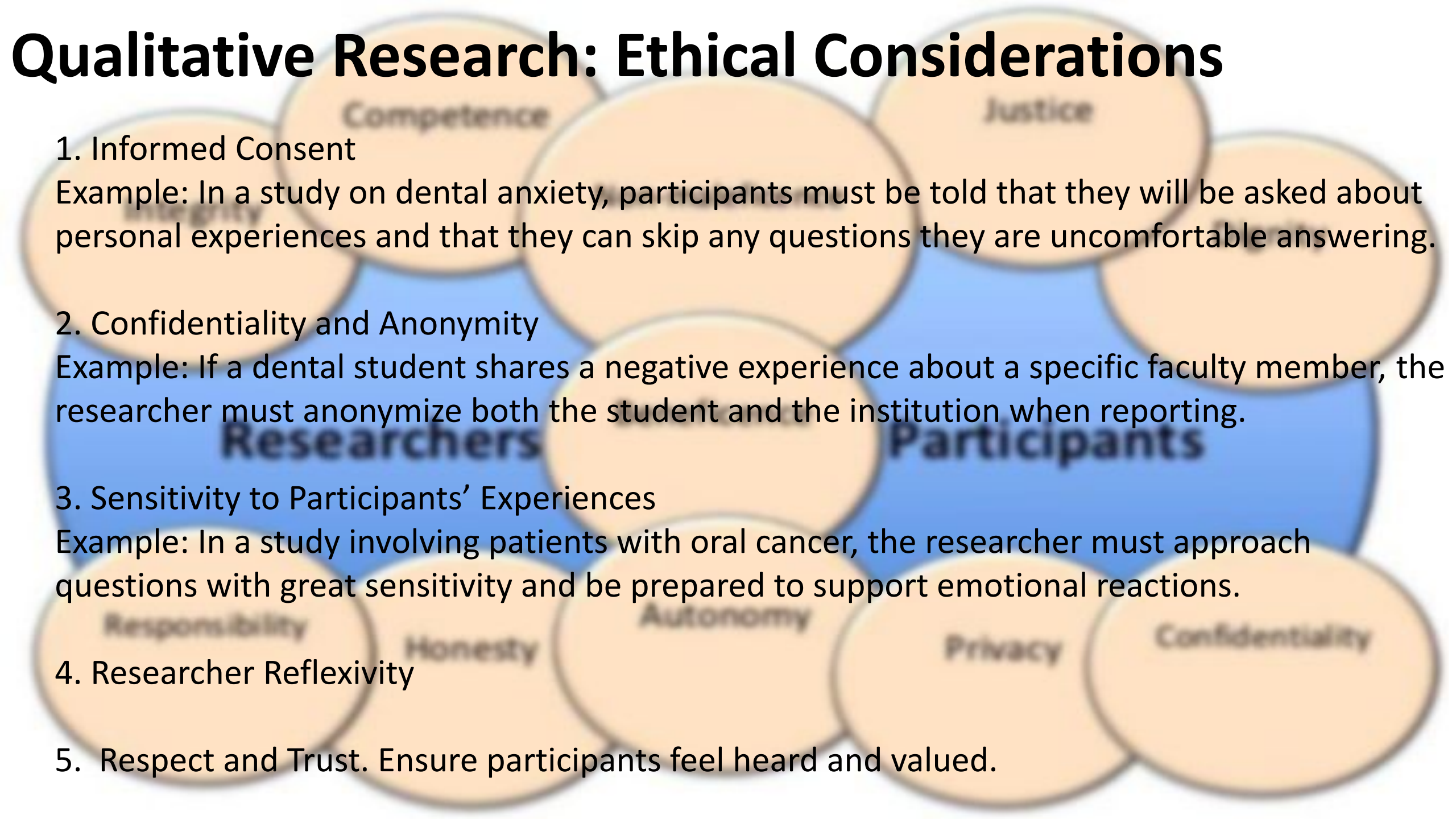
Allows for deeper meaning: Rather than seeking a single “truth,” qualitative research values multiple truths based on individual perspectives.

Example: Two patients with the same toothache might describe their experiences very differently

One may focus on physical pain.

Another may emphasize fear, shame, or frustration based on past encounters.

Qualitative Research: Ethical Considerations



1. Informed Consent

Example: In a study on dental anxiety, participants must be told that they will be asked about personal experiences and that they can skip any questions they are uncomfortable answering.

2. Confidentiality and Anonymity

Example: If a dental student shares a negative experience about a specific faculty member, the researcher must anonymize both the student and the institution when reporting.

3. Sensitivity to Participants' Experiences

Example: In a study involving patients with oral cancer, the researcher must approach questions with great sensitivity and be prepared to support emotional reactions.

4. Researcher Reflexivity

5. Respect and Trust. Ensure participants feel heard and valued.



- "The Effect of Fluoride Varnish on Caries Incidence in Children"
- → Quantitative – Experimental
- "Exploring Dental Anxiety Among Adolescents"
- Qualitative – Phenomenology/Grounded Theory
- "A Comparison of Oral Hygiene Practices Between Urban and Rural Populations"
- → Quantitative – Descriptive/Crossectional
- "Understanding How Dental Students Experience Peer Feedback During Clinical Training"
- → Qualitative – Ethnography/Grounded Theory/Phenomenology
- "The Relationship Between Sugar Consumption and Dental Caries in Teenagers"
- → Quantitative – Correlational
- "Impact of a Mobile App on Patients' Oral Hygiene Compliance"
- → Quantitative – Experimental/Quasi-Experimental
- "Perceptions of Professionalism Among Dental Students"
- → Qualitative – Phenomenology

- "Linking Sleep Quality and Temporomandibular Disorders in Young Adults"
- → Quantitative – Correlational
- "How Do Patients Perceive Empathy in Dentist-Patient Communication?"
- → Qualitative – Phenomenology/Grounded Theory
- "Effectiveness of Simulation-Based Learning on Clinical Skills in Dentistry Students"
- → Quantitative – Experimental
- "Barriers to Accessing Dental Care Among Low-Income Families"
- → Qualitative – Ethnography/Grounded Theory/Phenomenology
- "Dental Students' Reflections on Their First Clinical Encounter"
- → Qualitative – Narrative Inquiry
- "The Association Between Academic Stress and Oral Health Behaviors in University Students"
- → Quantitative – Correlational
- "Improving Gingival Health Through Motivational Interviewing"
- → Quantitative – Experimental/Quasi-Experimental

- "Study of Oral Health Beliefs in a Tribal Community"
- → Qualitative – Ethnography
- "Knowledge, Attitude, and Practice (KAP) Survey on Oral Cancer Awareness Among Smokers"
- → Quantitative – Descriptive (KAP Study)
- "Dentists' Experiences of Ethical Dilemmas in Private Practice"
- → Qualitative – Descriptive
- "Evaluating the Impact of Online Learning on Dental Students' Clinical Confidence"
- → Mixed Methods – Quantitative + Qualitative Interviews
- "Association Between Periodontal Disease and Diabetes Mellitus in Adults"
- → Quantitative – Correlational
- "Understanding the Cultural Beliefs Surrounding Tooth Loss in Older Adults"
- → Qualitative – Ethnography

- "Understanding the Experience of Burnout in Emergency Nurses
- Journal: Journal of Clinical Nursing (2021)
- Study Design: Qualitative descriptive study
- Sampling: Purposive sampling of emergency nurses
- Data Collection: Semi-structured interviews
- Analysis: Thematic analysis to identify patterns and themes related to burnout experiences

- “Exploring Patients’ Perceptions of Compassion in Nursing Care
- Journal: Journal of Advanced Nursing (2020)
- Study Design: Qualitative, Grounded theory/Phenomenology
- Sampling: Theoretical sampling of hospital patients
- Data Collection: In-depth interviews
- Analysis: Constant comparative method to generate a theory on compassion in nursing

- “Effect of Simulation-Based Education on the Clinical Skills of Medical Students
- Journal: Medical Education (2020)
- Design: Quantitative, Randomized Controlled Trial (RCT)
- Sample : 120 medical students randomized into intervention and control groups
- Data Collection: Pre-test/post-test using OSCE (Objective Structured Clinical Examination) scores
- Analysis: Independent t-test and ANOVA to compare means

- Medical Students' Experiences of Professionalism Dilemmas and the Role of Empathy
- Journal: BMJ Open (2022)
- Design: Qualitative, Interpretative Phenomenological Analysis (IPA)
- Sampling: Purposive sampling of final-year medical students
- Data Collection: Face-to-face interviews
- Analysis: IPA to explore lived experiences and meaning-making related to professionalism and empathy

- “Impact of Hand Hygiene Compliance on Hospital-Acquired Infection Rates
- Journal: American Journal of Infection Control (2019)
- Design: Quantitative, Quasi-experimental (pre-post intervention design)
- Sample: Observational data from multiple wards
- Data Collection: Hand hygiene audit scores and infection rate reports
- Analysis: Chi-square test and logistic regression

when waiting for the
boring lectures to end.



Thank You